

```
#include<iostream>

#include<fstream>

#include<cstring>

using namespace std;

class Book{

public:

    int id;

    char name[50];

    char auther[50];

    int quantity;

    void input(){

        cout<<"\nEnter book id: ";

        cin>>id;

        cout<<"Enter book name: ";

        cin>>name;

        cout<<"Enter book auther: ";

        cin>>auther;

        cout<<"Enter book quantity: ";

        cin>>quantity;

    }

    void display(){

        cout<<"\nBook ID: "<<id;

        cout<<"\nBook Name: "<<name;

        cout<<"\nBook Auther: "<<auther;

        cout<<"\nBook Quantity: "<<quantity;

        cout<<"\n-----";

    }

}
```

```
};
```

```
void addBook(){  
    Book b;  
    b.input();  
    ofstream file("Books.dat", ios::app | ios::binary);  
    file.write((char*)&b,sizeof(b));  
    file.close();  
  
    cout<<"\nBook added successfully !\n";  
}
```

```
void displayBook(){  
    Book b;  
    ifstream file("books.dat", ios::binary );  
  
    if(!file){  
        cout<<"\nNo record found !\n";  
        return;  
    }  
  
    while(file.read((char*)&b,sizeof(b))){  
        b.display();  
    }  
  
    file.close();  
}
```

```
void searchBook(){
```

```

int id;

cout<<"\nEnter book ID to search: ";

cin>>id;


Book b;

ifstream file("books.dat",ios::binary);


bool found = false;


while(file.read((char*)&b,sizeof(b))){
    if(b.id == id){
        cout<<"\nBook found:\n";

        b.display();

        found = true;

        break;
    }
}

if(!found){
    cout<<"\nBook not found !\n";
}

file.close();
}


void issueBook(){

int id;

cout<<"\nEnter book ID to issue: ";

cin>>id;


Book b;

fstream file("books.det", ios::in | ios::out | ios::binary);

```

```

while(file.read((char*)&b,sizeof(b))){
    if(b.id ==id){
        if(b.quantity > 0){
            b.quantity--;

            int pos = -1 *sizeof(b);
            file.seekp(pos, ios::cur);
            file.write((char*)&b,sizeof(b));

            cout<<"\nBook issued successfully !\n";
        }
        else{
            cout<<"\nBook out of stock !\n";
        }
        file.close();
        return;
    }
}

cout<<"\nBook not found !\n";
}

void returnBook(){
    int id;

    cout<<"\nEnter book ID to return: ";
    cin>>id;

    Book b;

    fstream file("books.det", ios::in | ios::out |ios::binary);

```

```

while(file.read((char*)&b,sizeof(b))){
    if(b.id == id){
        b.quantity++;

        int pos = -1 * sizeof(b);
        file.seekp(pos, ios::cur);
        file.write((char*)&b,sizeof(b));

        cout<<"\nBook returns successfully !\n";
        file.close();
        return;
    }
}
cout<<"\nBook not found !\n";
file.close();
}

```

```

void deleteBook(){
    int id;

    cout<<"\nEnter book ID to delete: ";
    cin>>id;

    Book b;

    ifstream fin("books.det", ios::binary);
    ofstream fout("temp.dat",ios::binary);

    bool deleted = false;

    while(fin.read((char*)&b,sizeof(b))){
        if(b.id != id){

```

```

        fout.write((char*)&b,sizeof(b));
    }
    else{
        deleted = true;
    }
}

fin.close();
fout.close();

remove("books.dat");
rename("temp.dat","books.dat");

if(deleted){
    cout<<"\nBook deleted successfully !\n";
}
else{
    cout<<"\nBook not found !\n";
}
}

int main(){
    int choice;

    while(true){
        cout<<"\n===== Library Management System =====";
        cout<<"\n1. Add Book";
        cout<<"\n2. Display Books";
        cout<<"\n3. Search Book";
        cout<<"\n4. Issue Book";
    }
}

```

```
cout<<"\n5. Return Book";

cout<<"\n6. Delete Book";

cout<<"\n7. Exit";

cout<<"\nEnter choice: ";

cin>>choice;


switch(choice){

    case 1:

        addBook();

        break;

    case 2:

        displayBook();

        break;

    case 3:

        searchBook();

        break;

    case 4:

        issueBook();

        break;

    case 5:

        returnBook();

        break;

    case 6:

        deleteBook();

        break;

    case 7:

        cout<<"\nThank You !\n";

        return 0;

    default:

        cout<<"\nInvalid choice !";
```

}

}

}